

Simple Harmonic Motion: Sample Problems

A **0.42-kg** block is attached to the end of a horizontal ideal spring and rests on a frictionless surface. The block is pulled so that the spring stretches by **2.1 cm** relative to its unstrained length. When the block is released, it moves with an acceleration **of 9.0 m/s²**. What is the spring constant of the spring?

180 N/m

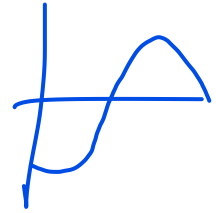
Simple Harmonic Motion: Sample Problems

Energy calculations.

For the simple harmonic oscillation where $k = 19.6$

N/m , $A = 0.100 \text{ m}$, $x = -(0.100 \text{ m}) \cos 8.08t$, and $v =$

- $(0.808 \text{ m/s}) \sin 8.08t$, determine (a) the total energy, (b) the kinetic and potential energies as a function of time, (c) the velocity when the mass is 0.050 m from equilibrium, (d) the kinetic and potential energies at half amplitude ($x = \pm A/2$).



Simple Harmonic Motion: Sample Problems

$$\text{a. } E = \frac{1}{2}kA^2 = \frac{1}{2} \cdot 19.6 \text{ N/m} \cdot (0.100 \text{ m})^2 = 9.80 \times 10^{-2} \text{ J}.$$

$$\text{b. } U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2 \omega t = (9.80 \times 10^{-2} \text{ J}) \cos^2 8.08t,$$

$$K = E - U = (9.80 \times 10^{-2} \text{ J}) \sin^2 8.08t.$$

$$\text{c. } K = E - U, \quad \frac{1}{2}mv^2 = \frac{1}{2}kA^2 - \frac{1}{2}kx^2,$$

$$v = \sqrt{\frac{k}{m}(A^2 - x^2)} = \omega \sqrt{A^2 - x^2}$$

$$= 8.08 \text{ Hz} \cdot \sqrt{(0.100 \text{ m})^2 - (0.050 \text{ m})^2} = 0.70 \text{ m/s}.$$

$$\text{d. } U = \frac{1}{2}kx^2 = \frac{1}{2}k\left(\frac{A}{2}\right)^2 = \frac{1}{4}E = 2.5 \times 10^{-2} \text{ J},$$

$$E = K - U = 7.3 \times 10^{-2} \text{ J}.$$

Simple Harmonic Motion: Sample Problems

A 500 g block on a spring is pulled a distance of 20 cm and released. The subsequent oscillations are measured to have a period of 0.80 s. At what position (or positions) is the speed of the block 1.0 m/s?

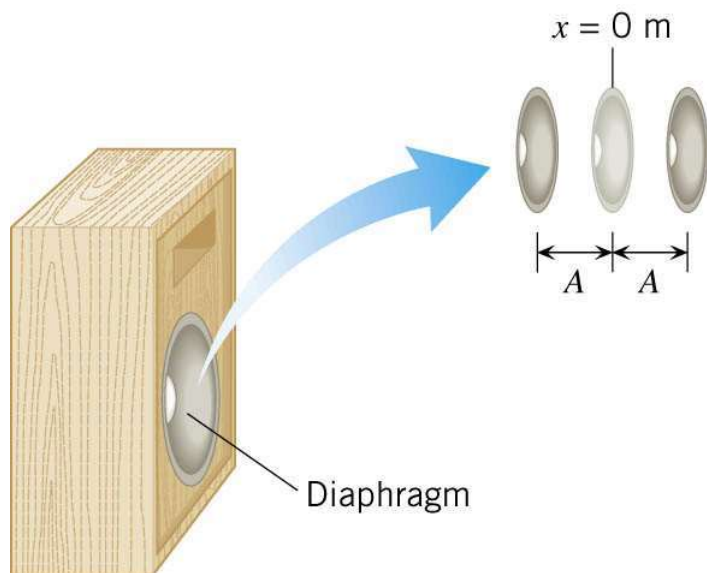
$$T = 0.80 \text{ s} \quad \text{so} \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{(0.80 \text{ s})} = 7.85 \text{ rad/s}$$

$$v = \sqrt{\frac{k}{m}(A^2 - x^2)} = \omega\sqrt{A^2 - x^2}$$

$$x = \pm\sqrt{A^2 - \left(\frac{v}{\omega}\right)^2} = \pm\sqrt{(0.20 \text{ m})^2 - \left(\frac{(1.0 \text{ m/s})}{(7.85 \text{ rad/s})}\right)^2} = \pm 0.154 \text{ m} = \pm 15.4 \text{ cm}$$

Simple Harmonic Motion: Sample Problems

The diaphragm of a loudspeaker moves back and forth in simple harmonic motion to create sound. The frequency of the motion is $f = 1.0$ kHz and the amplitude is $A = 0.20$ mm.



- (a) What is the maximum speed of the diaphragm?
- (b) Where in the motion does this maximum speed occur?

Simple Harmonic Motion: Sample Problems

(a)

$$v_{\max} = A\omega = A(2\pi f) = (0.20 \times 10^{-3} \text{ m})(2\pi)(1.0 \times 10^3 \text{ Hz}) = \boxed{1.3 \text{ m/s}}$$

(b) The speed of the diaphragm is zero when the diaphragm momentarily comes to rest at either end of its motion: $x = +A$ and $x = -A$. Its maximum speed occurs midway between these two positions, or at $x = 0$ m.

Simple Harmonic Motion: Sample Problems

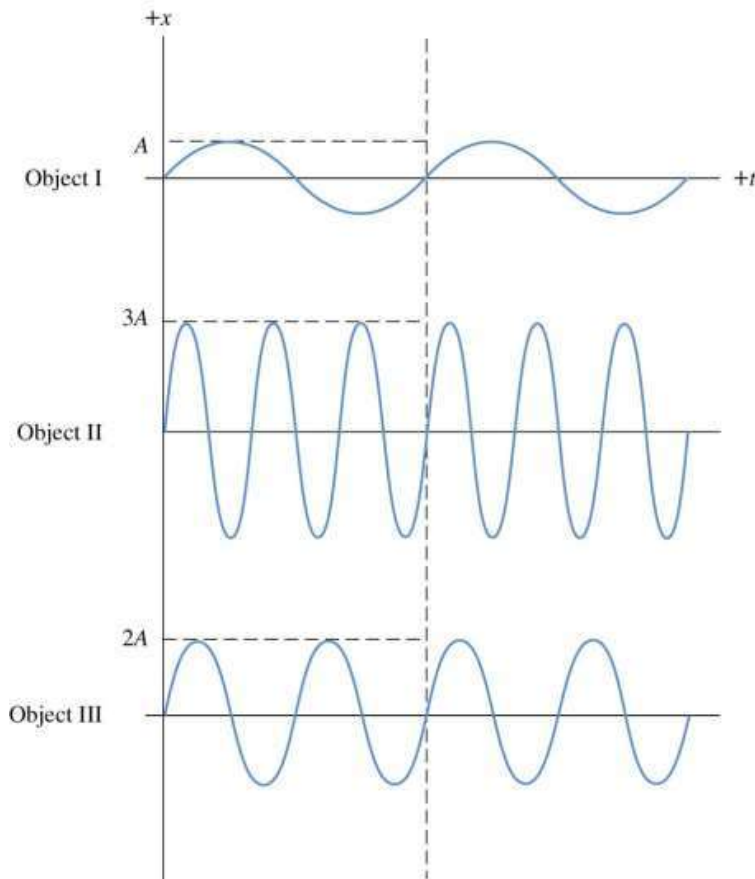
What is the oscillation period of an FM radio station that broadcasts at 100 MHz?

$$f = 100 \text{ MHz} = 1.0 \times 10^8 \text{ Hz}$$

$$T = 1/f = \frac{1}{1.0 \times 10^8 \text{ Hz}} = 1.0 \times 10^{-8} \text{ s} = 10 \text{ ns}$$

Note that $1/\text{Hz} = \text{s}$

Simple Harmonic Motion: Sample Problems



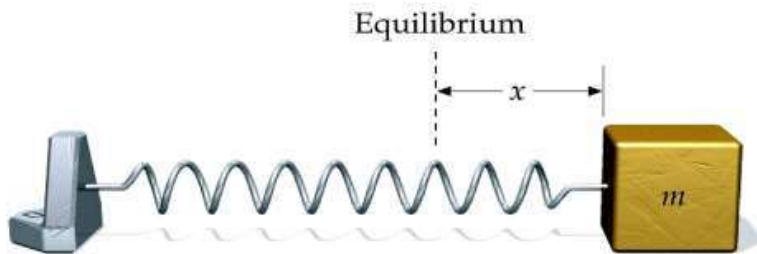
The drawing shows plots of the displacement x versus the time t for three objects undergoing simple harmonic motion. Which object, I, II, or III, has the greatest maximum velocity?

II

Simple Harmonic Motion: Sample Problems

A 2.00 kg block is attached to a spring as shown. The force constant of the spring is $k = 196 \text{ N/m}$. The block is held a distance of 5.00 cm from equilibrium and released at $t = 0$.

- (a) Find the angular frequency ω , the frequency f , and the period T .
- (b) Write an equation for x vs. time.



Ref: google image

Simple Harmonic Motion: Sample Problems

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{(196 \text{ N/m})}{(2.00 \text{ kg})}} = 9.90 \text{ rad/s}$$

$$f = \frac{\omega}{2\pi} = \frac{(9.90 \text{ rad/s})}{2\pi} = 1.58 \text{ Hz}$$

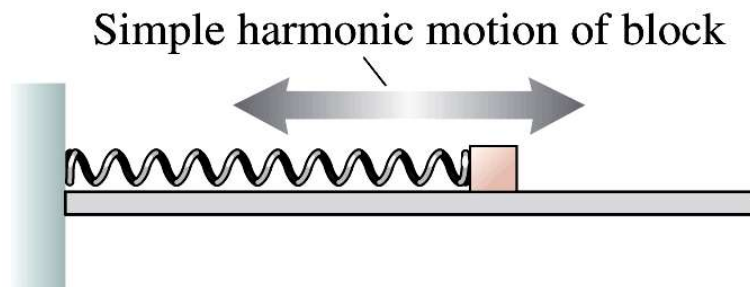
$$T = 1/f = 0.635 \text{ s} \quad A = 5.00 \text{ cm} \text{ and } \delta = 0$$

$$x = (5.00 \text{ cm}) \cos[(9.90 \text{ rad/s})t]$$

Simple Harmonic Motion: Sample Problems

An air-track glider is attached to a spring, pulled 20 cm to the right, and released at $t=0$. It makes 15 complete oscillations in 10 s.

- What is the period of oscillation?
- What is the object's maximum speed?
- What is its position and velocity at $t=0.80$ s?



Ref: google image

Simple Harmonic Motion: Sample Problems

$$f = \frac{15 \text{ oscillations}}{10 \text{ s}}$$

$$= 1.5 \text{ oscillations/s} = 1.5 \text{ Hz}$$

$$T = 1/f = 0.667 \text{ s}$$

$$v_{\max} = \frac{2\pi A}{T} = \frac{2\pi(0.20 \text{ m})}{(0.667 \text{ s})} = 1.88 \text{ m/s}$$

$$x = A \cos \frac{2\pi t}{T} = (0.20 \text{ m}) \cos \frac{2\pi(0.80 \text{ s})}{(0.667 \text{ s})} = 0.062 \text{ m} = 6.2 \text{ cm}$$

$$v = -v_{\max} \sin \frac{2\pi t}{T} = -(1.88 \text{ m/s}) \sin \frac{2\pi(0.80 \text{ s})}{(0.667 \text{ s})} = -1.79 \text{ m/s}$$

Simple Harmonic Motion: Sample Problems

A mass, oscillating in simple harmonic motion, starts at $x = A$ and has period T .

At what time, as a fraction of T , does the mass first pass through $x = \frac{1}{2}A$?

$$x = \frac{1}{2} A = A \cos \frac{2\pi t}{T}$$

$$t = \frac{T}{2\pi} \cos^{-1} \left(\frac{1}{2} \right) = \frac{T}{2\pi} \frac{\pi}{3} = \frac{1}{6} T$$